

Earth's Circumference

Measurements, calculations and the unfolded plane

The Enoch Project — Phase 9, report on the scale test and circumference

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This report presents four physical latitude measurements taken with the Google Earth grid, combined with the Lake Pontchartrain Causeway as a fixed reference span, and calculates Earth's circumference based on the radius that Google Earth itself displays (10,001.47 km). The result is compared with direct observations of the Sun's path at the tropics and the Arctic Circle.

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1. Four physical latitude measurements

Four data points collected using the Google Earth grid. The grid is locked to lat/long and is used as the measurement tool. All measurements run north-south along the same line of longitude.

Segment	Span	Total km	km / degree
Equator → 10°N (Africa)	10°	1,105.93	110.593
30°N → 35°N (USA)	5°	554.60	110.920
60°N → 65°N (Northern Europe)	5°	557.36	111.472
60°N → 70°N	10°	1,115.25	111.525

Derived from 60°N→70°N minus 60°N→65°N:

$$65^{\circ}\text{N} \rightarrow 70^{\circ}\text{N} = 1,115.25 - 557.36 = 557.89 \text{ km}$$

$$557.89 / 5 = 111.58 \text{ km / degree}$$

Difference between two 10-degree segments:

Segment	Meters
Equator → 10°N	1,105,930 m
60°N → 70°N	1,115,250 m
Difference	9,320 m

Explanation:

As we move north, each degree of latitude becomes gradually longer. From the equator up to 10°N, ten degrees equal 1,105.93 km. From 60°N up to 70°N, ten degrees equal 1,115.25 km. That is nearly 10 km more for ten degrees farther north. The difference is 9,320 m.

This pattern — distances between latitudes increasing toward the north — matches what Enoch's model shows when projected flat: the further from the centre (the North Pole), the larger the rings become.

2. Lake Pontchartrain Causeway — reference measurement

The Lake Pontchartrain Causeway is the world's longest straight bridge over water. It is built of 2,250 identical concrete spans, making it a physically fixed measuring tape against which we can calibrate.

Endpoints from the grid:

Point	Latitude	Longitude
South start	30°01'13.45"N	90°06'41.76"W
North end	30°21'57.57"N	90°06'02.88"W

Measurement via the grid:

$$96 \text{ squares} \times 400 \text{ m} = 38,400 \text{ m}$$

$$10 \text{ squares along the bridge} = 37.01 \text{ km} = 37,010 \text{ m}$$

$$\text{Deviation per 10 squares} = 90 \text{ m}$$

Factory specification:

$$2,250 \text{ spans} \times 56 \text{ ft} \times 0.3048 \text{ m} = 38,404.8 \text{ m}$$

Summary, Pontchartrain:

Source	Length
Grid 96 × 400 m	38,400.0 m
Factory 2,250 × 56 ft	38,404.8 m
Difference	4.8 m

The grid and the factory specification agree within 5 metres over 38 km. That means the Google Earth grid is reliable as a measurement tool for short distances on the same line of latitude.

1 degree from 30°N to 35°N:

$$\text{Measured GE distance: } 554.60 \text{ km}$$

$$554.60 \text{ km} / 5 \text{ degrees} = 110.92 \text{ km} / \text{degree}$$

$$554,600 \text{ m} / 40 \text{ m} = 13,865 \text{ units}$$

3. Calculating Earth's circumference with the GE radius

Google Earth displays Earth's radius as 10,001.47 km. We use that value directly and substitute into the standard circle formula.

Input:

$$R = 10,001.47 \text{ km} = 10,001,470 \text{ m}$$

Formula:

$$\text{Circumference} = 2 \times \pi \times R$$

Value of π :

$$\pi = 3.141\,592\,653\,589\,793$$

$$2\pi = 6.283\,185\,307\,179\,586$$

Substituted:

$$\text{Circumference} = 6.283\,185 \times 10,001.47$$

$$\text{Circumference} = 62,841.089 \text{ km}$$

$$\text{Circumference} = 62,841,089 \text{ m}$$

Diameter:

$$D = 2 \times R = 20,002.94 \text{ km}$$

Summary:

Quantity	Value
GE radius	10,001.47 km
Computed diameter	20,002.94 km
Computed circumference	62,841.09 km
Official circumference (WGS84)	40,075.02 km
Difference	22,766.07 km

Explanation:

When the radius is 10,001.47 km the circumference must be 62,841.09 km — it follows directly from π . The official circumference of 40,075 km belongs to an entirely different radius (6,378 km). The two figures cannot both be correct at the same time.

Reverse calculation from the official circumference:

$$R = \text{Circumference} / (2\pi)$$

$$R = 40,075 / 6.283\ 185 = 6,378.14\text{ km}$$

The official circumference yields a radius of 6,378 km — not 10,001 km as shown in Google Earth's own radius reading. This is a contradiction that cannot be resolved within the spherical model.

4. The plane unfolded — half the circumference as a new diameter

If you take a sphere and cut it in half, then lay one hemisphere flat, then half of the original circumference becomes the new diameter of the flat surface. That is the surface we are standing on.

Step 1 — Half the circumference:

$$\text{Circumference} = 62,841.089 \text{ km}$$

$$\text{Half the circumference} = 62,841.089 / 2 = 31,420.545 \text{ km}$$

Step 2 — Half the circumference becomes the new diameter:

$$\text{New diameter } d = 31,420.545 \text{ km}$$

$$\text{New radius } r = d / 2 = 15,710.272 \text{ km}$$

Step 3 — Circumference of the unfolded surface:

$$\text{New circumference} = \pi \times d$$

$$\text{New circumference} = 3.141\,592\,653 \times 31,420.545$$

$$\text{New circumference} = 98,710.552 \text{ km}$$

Summary — the full chain:

Step	Value
GE radius R	10,001.47 km
Sphere diameter 2R	20,002.94 km
Sphere circumference $2\pi R$	62,841.09 km
Half the circumference	31,420.545 km
New diameter on unfolded plane	31,420.545 km
New radius on unfolded plane	15,710.27 km
New circumference on unfolded plane ($\pi \times d$)	98,710.55 km
Official circumference (WGS84)	40,075.02 km

What the numbers show:

When the sphere is cut in half and the surface unfolded, the new circumference becomes 98,710.55 km. That is nearly 2.5 times the official 40,075 km. The ratio is exactly $\pi/2$ — the unfolded plane's circumference is $\pi/2$ times larger than the spherical circumference.

Ratios between the three circumferences:

Comparison	Ratio
Unfolded plane / Sphere circumference	1.5708 ($\pi/2$)
Unfolded plane / Official circumference	2.463
Sphere circumference / Official circumference	1.568

Consequence: If Google Earth's radius of 10,001.47 km is correct, and the surface is in fact flat as the Sun's path indicates, then the outer circumference (the Antarctic ring enclosing the continents) is 98,710.55 km — not 40,075 km.

5. The Sun's path — direct observations

Three independent observations that tell us where the Sun is in the sky at the tropics and the Arctic Circle.

5.1 El día sin sombras — Antofagasta, Chile

21 December at 13:39 local time in Antofagasta, at the Tropic of Capricorn monument. Vertically standing objects lose their shadows. The Sun is directly overhead — at zenith.

[Video witness: El día sin sombras — Antofagasta, Chile \(Claux.7, 26 December 2020, 11:56\)](#)

5.2 The 45° rule

When the Sun stands at zenith (90° up) over a tropic, it stands 45° up as seen from the equator. The equator lies geometrically halfway between the two tropics. Half angle — because the Sun's position over the surface produces symmetric shadow geometry from Cancer to Capricorn.

$$\tan(45^\circ) = 1$$

Sun's height = horizontal distance to the zenith point

5.3 Grímsey, Iceland

18 December at 12 local time, Grímsey at 66°33'N (on the Arctic Circle itself): the Sun is 0° due south — right on the horizon. The Sun's zenith point lies at the same moment on the Tropic of Capricorn 23°27'S.

Angular distance between observer and zenith point:

$$66^\circ 33' \text{N} + 23^\circ 27' \text{S} = 90^\circ 00'$$

When the distance along the surface corresponds to 90° of angular separation, the Sun is on the horizon. This holds all the way around the Arctic Circle at winter solstice, and the opposite at summer solstice (midnight sun).

5.4 Connection

The three observations connect geometrically:

Location	Sun's height	Angle to zenith
Antofagasta (Capricorn), 21 Dec 13:39	90° (zenith)	0°
Equator, same day	45°	23°27'
Grímsey (66°33'N), same day at 12	0°	90°

The Sun's height above the horizon decreases linearly with angular distance from the zenith point. This is consistent with a flat geometry where the Sun stands above the surface at a certain height, and its rays meet the ground at an angle equal to the horizontal distance to the point directly below it.

6. Witnesses along the Arctic Circle

Midnight sun at summer solstice is documented all the way around the Arctic Circle — from Alaska in the west, through Iceland and Scandinavia, to Russia and Siberia in the east. These places sit at 66°33'N or further north, and the Sun stays above the horizon at midnight on or around 21 June.

Location	Latitude	Longitude	Country
Toolik Field Station	68°38'N	149°36'W	USA (Alaska)
Coldfoot	67°15'N	150°10'W	USA (Alaska)
Fort Yukon	66°34'N	145°15'W	USA (Alaska)
Grímsey	66°33'N	18°01'W	Iceland
Bodø	67°17'N	14°23'E	Norway
Inari	68°54'N	27°02'E	Finland
Murmansk	68°58'N	33°05'E	Russia
Norilsk	69°20'N	88°13'E	Russia
Tiksi	71°39'N	128°52'E	Russia (Sakha)

Coverage on five of six longitude sectors around the circle. Central Canada is missing from the list because its Arctic regions are sparsely documented for tourism, not because the phenomenon is any different there.

These witness accounts are independent of Google Earth, independent of any particular cosmological model, and independent of one another. They show that the Sun's path around the Arctic Circle is an actual, observable reality.

7. Conclusion

Summary of what we have measured and calculated together:

Two contradictory numbers for Earth's size

Google Earth shows the radius as 10,001.47 km. The official circumference (WGS84) is 40,075.02 km, which only makes sense with a radius of 6,378.14 km. The two values cannot both be correct. When we use Google Earth's own radius in the circle formula, the circumference becomes 62,841.09 km — more than 22,700 km greater than the official figure.

When the surface is unfolded, the circumference becomes 98,710.55 km

If the sphere is cut in two and half the circumference becomes the new diameter of an unfolded plane, the new circumference is 98,710.55 km — $\pi/2$ times larger than the spherical circumference, and nearly 2.5 times larger than the official figure. That is the outer ring enclosing the continents.

Latitude spacing increases toward the north

We measure the distance between degrees of latitude to be 110.59 km/degree at the equator, and 111.53 km/degree at 60°–70°N. A difference of nearly 10 km when comparing ten-degree segments. This pattern — rings that grow larger toward the north — matches Enoch's model projected flat.

The Sun's path is consistent with flat geometry

The Antofagasta video from 21 December directly shows the Sun at zenith over the Tropic of Capricorn. The 45° rule and the Grímsey observation follow the same angle arithmetic: Sun's height = 90° minus angular distance to the zenith point. Midnight-sun witnesses all the way around the Arctic Circle confirm that the Sun moves in circles over the plane.

Premise

No one has measured Earth's circumference directly. The official 40,075 km is a derived value from an assumed radius and assumed geometry. What we have here are direct measurements with the grid, a physically fixed bridge segment, observed Sun positions at the tropics, and witnesses along the Arctic Circle. The numbers agree with each other and with Enoch's model. The official value does not agree with Google Earth's own radius reading.

8. Sources and evidence

External sources

1. Lake Pontchartrain Causeway — factory specification

2,250 spans $\times$ 56 ft = 38,404.8 m. World's longest straight bridge over water.

https://en.wikipedia.org/wiki/Lake_Pontchartrain_Causeway

2. Earth radius — WGS84

Official equatorial circumference 40,075.017 km and equatorial radius 6,378.137 km. Used as a reference for the official figures.

https://en.wikipedia.org/wiki/Earth_radius

3. El día sin sombras — Antofagasta

Direct video witness at the Tropic of Capricorn, 21 December at 13:39. Claux.7, published 26 December 2020.

<https://www.youtube.com/watch?v=NROhPbKhclo>

4. Grímsey — summer solstice festival

Visit Akureyri documents the annual festival on the Arctic Circle at 66°33'N.

<https://www.visitakureyri.is/en/see-and-do/events-festivals/summer-solstice-festival-in-grimsey-june>

5. Bodø — midnight sun from June to mid-July

Visit Bodø, official documentation.

<https://visitbodo.com/en/topic/midnight-sun/>

6. Murmansk — world's largest city above the Arctic Circle

Wikipedia entry, midnight-sun phenomenon.

<https://en.wikipedia.org/wiki/Murmansk>

7. Tiksi — midnight sun from 1 May to 11 August

Wikipedia, Russian Arctic seaport at 71°39'N.

<https://en.wikipedia.org/wiki/Tiksi>

8. Toolik Field Station — time-lapse 15-16 June 2019

University of Alaska Fairbanks, midnight sun observed directly.

<https://www.youtube.com/watch?v=Sl6DbRoX9X4>

Internal reports (workspace)

9. Measuring tape v1 report

Four physical segments on three continents. Pontchartrain, 90 Mile Straight, Trans-Australian Railway, Highway 10 Saudi Arabia. Mean absolute deviation 0.041%.

<workspace:/maalband-v1-rapport.md>

10. Tropic witnesses

Catequilla in Ecuador, Aswan in Egypt and other documented zenith observations at the tropics.

<workspace:/vendekretser-vitnesbyrd.pdf>

11. Midnight-sun witnesses

Complete list of 15 documented midnight-sun locations around the Arctic Circle, sorted by longitude.

<workspace:/midnattssol-vitnesbyrd.md>

12. Meridian-gold report

Analysis of meridian transits and gold-pole observations.

<workspace:/meridian-gull-rapport.pdf>

13. Punta Arenas report

Documentation of the southern zenith test at 53°S, including HMS Challenger 1876 and Wales 1774 historical ship logs.

[workspace:/punta-arenas-rapport.pdf](#)

14. Equator-zenith report

The Catequilla monument and other equator-zenith observations.

[workspace:/ekvator-zenith-rapport-v1.md](#)